

Lecture 33: Nov 10

Last time

- Midterm exam 2

Today

- Multiple Random Variables (Chapter 4)

Notes

- Normal distribution is useful in many practical settings. E.g. measurement error.
- Plays an important role in *sampling distributions* in *large samples*, since the Central Limit Theorem says that the mean (and also the sum) of independent identically distributed random variables are approximately normal
- There are many important distributions that can be derived from functions of normal random variables (e.g., χ^2 , t , F). We will briefly present the pdf's and sample spaces of these distributions.

χ^2 distribution If $Z \sim N(0,1)$, then $X = Z^2$ has the χ^2 distribution with 1 degree of freedom. More generally, we have the χ^2 distribution with v degrees of freedom with pdf:

$$f(x) = \frac{(x/2)^{\frac{v}{2}-1} e^{-x/2}}{2\Gamma(v/2)}, \quad x > 0$$

where $\Gamma(a)$ is the complete gamma function,

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx$$

The $\chi^2(v)$ distribution is a special case of the gamma distribution, so it is easier to derive its properties from the gamma.

Facts about the Gamma function

- $\Gamma(a+1) = a\Gamma(a), a > 0$
- $\Gamma(1) = 1$
- $\Gamma(n) = (n-1)!$
- $\Gamma(1/2) = \sqrt{\pi}$

Gamma distribution Notation: $Y \sim \text{Gamma}(a, \lambda)$.

- pdf:

$$f(y) = \frac{\lambda e^{-\lambda y} (\lambda y)^{a-1}}{\Gamma(a)}, \quad y \geq 0$$

where $\Gamma(a)$ is the gamma function,

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx$$

- cdf: In general, there is no closed form, unless a is an integer.
- moments:

$$\begin{aligned} E(Y) &= a/\lambda \\ Var(Y) &= a/\lambda^2 \end{aligned}$$

- MGF:

$$M_Y(t) = \left(\frac{1}{1 - t/\lambda} \right)^a, \quad t < \lambda$$

Another parameterization Same as the exponential distribution, we can let $\beta = \frac{1}{\lambda}$, then we have

- pdf:

$$f(y) = \frac{y^{a-1} e^{-y/\beta}}{\Gamma(a) \beta^a}, \quad y \geq 0$$

- moments:

$$\begin{aligned} EX &= a\beta \\ Var(X) &= a\beta^2 \end{aligned}$$

- MGF:

$$M_Y(t) = \left(\frac{1}{1 - t\beta} \right)^a, \quad t < \frac{1}{\beta}$$

Notes:

- The special case $a = 1$ corresponds to an *exponential*(λ)
- The parameter a is known as the *shape parameter*, since it most influences the peakedness of the distribution.
- The parameter β is called the *scale parameter* since most of its influence is on the spread of the distribution.
- The special case $Gamma(a = n/2, \lambda = 1/2)$, for integer n , corresponds to the χ_n^2 distribution with n degrees of freedom.
- The gamma distribution can be derived as the sum of a (the shape parameter) independent *exponential*(λ) distributions.

Beta distribution Notation: $Y \sim \text{Beta}(a, b)$.

- Sample space: $[0, 1]$
- pdf:

$$f(y) = \frac{y^{a-1}(1-y)^{b-1}}{B(a, b)}, \quad 0 \leq y \leq 1$$

where $B(a, b)$ is the Beta function,

$$B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1}dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

and $\Gamma(a)$ is the gamma function. Note that if a and b are integers, then $B(a, b)$ can be calculated in closed form.

- cdf: In general, there is no closed form, except if a and b are integers.
- moments:

$$EY = \frac{a}{a+b}$$
$$Var(Y) = \frac{ab}{(a+b)^2(a+b+1)}$$

The beta distribution is very flexible, and can take a wide variety of shapes by varying its parameters.

- Special case: $\text{Beta}(1, 1) = U(0, 1)$.

Omitted distributions: Weibull distribution and Cauchy distribution.

Location and Scale families

Let Z be a continuous random variable with pdf $f(z)$. Define the class of rvs

$$X_{\mu, \sigma} = \sigma Z + \mu, \quad \mu \in \mathbb{R}, \sigma > 0$$

Then

1. $X_{\mu, \sigma}$ has pdf

$$f_{\mu, \sigma}(x) = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right)$$

- 2.

$$E(X) = \sigma E(Z) + \mu, \quad Var(X) = \sigma^2 Var(Z)$$

3. The variable $Z = X_{0,1}$ is called the *generator* and is a member of the class.

Location families and scale families

- The family of pdfs $f_{\mu,\sigma}(x)$ is called a *location-scale* family where μ is called the *location parameter*, and σ is called the *scale parameter*.

- The family of pdfs

$$f_{\mu,1}(x) = f(x - \mu)$$

with $\sigma = 1$ is called a *location* family.

- The family of pdfs

$$f_{0,\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right)$$

with $\mu = 0$ is called a *scale* family.

Example (Exponential location family) Let $f(x) = e^{-x}$, $x \geq 0$, and $f(x) = 0$, $x < 0$. To form a location family we replace x with $x - \mu$ to obtain

$$\begin{aligned} f(x|\mu) &= \begin{cases} e^{-(x-\mu)} & x - \mu \geq 0 \\ 0 & x - \mu < 0 \end{cases} \\ &= \begin{cases} e^{-(x-\mu)} & x \geq \mu \\ 0 & x < \mu \end{cases} \end{aligned}$$

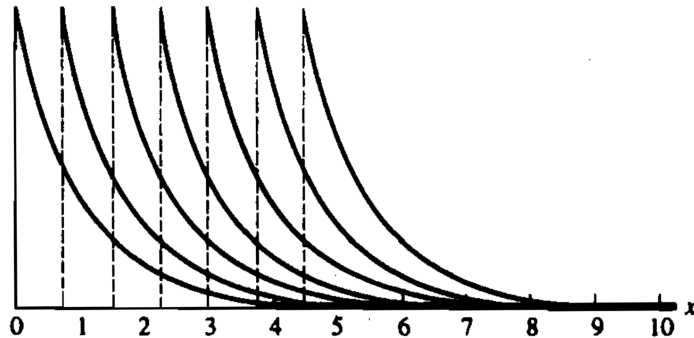


Figure 3.5.2. Exponential location densities

Figure 32.1: Figure 3.5.2. Exponential location densities.

As shown in the above graph, the densities are shifted. Now the positive part of the density starts at μ rather than at 0. If X measures time, then μ might be restricted to be nonnegative so that X will be positive with probability 1 for every value of μ . In this type of model, where μ denotes a bound on the range of X , μ is sometimes called a *threshold parameter*.

The effect of introducing the scale parameter σ is either to stretch ($\sigma > 1$) or to contract ($\sigma < 1$) the graph of $f(x)$ while still maintaining the same basic shape of the graph. This is illustrated in the Figure below.

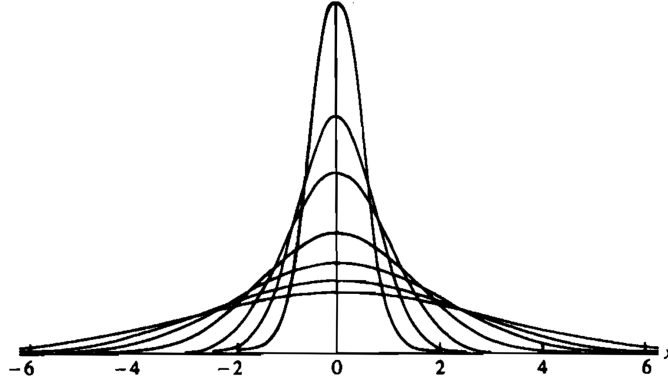


Figure 32.2: Figure 3.5.3. Members of the same scale family

Exponential Families A family of pdfs or pmfs with vector parameter $\boldsymbol{\theta}$ is called an *exponential family* if it can be expressed as

$$f(x|\boldsymbol{\theta}) = h(x)c(\boldsymbol{\theta})\exp\left(\sum_{j=1}^k w_j(\boldsymbol{\theta})t_j(x)\right), \quad x \in S \subset \mathbb{R} \quad (1)$$

where S is not defined in terms of $\boldsymbol{\theta}$, $h(x)$, $c(\boldsymbol{\theta}) \geq 0$ and the functions are just functions of the parameters specified; i.e. h is free of $\boldsymbol{\theta}$, $c(\boldsymbol{\theta})$ is free of x , etc...

Examples:

- One-dimensional: Exponential, Poisson
- Two-dimensional: Gaussian

Exponential family parameterizations are unique except for multiplying constant factors.

Example: Gaussian Let $f(x|\mu, \sigma^2)$ be the $n(\mu, \sigma^2)$ family of pdfs, where $\boldsymbol{\theta} = (\mu, \sigma)$. Then

$$\begin{aligned} f(x|\mu, \sigma^2) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) \exp\left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2}\right) \end{aligned}$$

Thus

$$\begin{aligned} h(x) &= \frac{1}{\sqrt{2\pi}} & c(\mu, \sigma) &= \frac{1}{\sigma} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) \\ w_1(\mu, \sigma) &= -\frac{1}{2\sigma^2} & w_2(\mu, \sigma) &= \frac{\mu}{\sigma^2} \\ t_1(x) &= x^2 & t_2(x) &= x \end{aligned}$$

The parameter space is $(\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$.

Example: Binomial Let $f(x|p)$ be the *binomial* (n, p) , $0 < p < 1$ family of pmfs.

$$\begin{aligned} f(x|p) &= \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} (1-p)^n \left[\frac{p}{1-p} \right]^x \\ &= \binom{n}{x} (1-p)^n \exp \left[\log \left(\frac{p}{1-p} \right) x \right] \end{aligned}$$

Thus,

$$\begin{aligned} h(x) &= \binom{n}{x}, \quad x = 0, \dots, n \quad w_1(p) = \log \left(\frac{p}{1-p} \right) \\ c(p) &= (1-p)^n, \quad 0 < p < 1 \quad t_1(x) = x \end{aligned}$$

Note that this works when p is considered the parameter, while n is fixed. Also, p cannot be 0 or 1. Otherwise, the range changes.

More examples The following distributions belong to Exponential families:

- Continuous: exponential, Gaussian, gamma, beta, χ^2
- Discrete: Poisson, geometric, binomial (fixed # trials), negative binomial (fixed # successes)

The following distributions do not belong to exponential families:

- Continuous: t , F , uniform
For example, $X \sim U(0, \theta)$, $f_X(x) = \theta^{-1} I_{0 < x < \theta}$
- Discrete: uniform, hypergeometric

Theorem If X is a random variable with pdf or pmf of the form [1](#), then

$$\begin{aligned} E \left(\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right) &= -\frac{\partial}{\partial \theta_j} \log c(\boldsymbol{\theta}) \\ Var \left(\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right) &= -\frac{\partial^2}{\partial \theta_j^2} \log c(\boldsymbol{\theta}) - E \left(\sum_{i=1}^k \frac{\partial^2 w_i(\boldsymbol{\theta})}{\partial \theta_j^2} t_i(X) \right). \end{aligned}$$

Although these equations may look formidable, when applied to specific cases they can work out quite nicely. Their advantage is that we can replace integration or summation by differentiation, which is often more straightforward.

Example (Normal exponential family) Let $f(x|\mu, \sigma^2)$ be the $n(\mu, \sigma^2)$ family of pdfs, where $\boldsymbol{\theta} = (\mu, \sigma)$, $-\infty < \mu < \infty, \sigma > 0$. Then

$$\begin{aligned} f(x|\mu, \sigma^2) &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x-\mu)^2}{2\sigma^2} \right) \\ &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{\mu^2}{2\sigma^2} \right) \exp \left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} \right) \end{aligned}$$

Define

$$\theta_1 = \frac{1}{\sigma^2} > 0, \quad \theta_2 = \frac{\mu}{\sigma^2} \in \mathbb{R}$$

Then

$$f_X(x) = \frac{\sqrt{\theta_1}}{\sqrt{2\pi}} \exp\left(-\frac{\theta_2^2}{2\theta_1}\right) \exp\left(-\theta_1 \frac{x^2}{2} + \theta_2 x\right)$$

and

$$\begin{aligned} h(x) &= 1 \text{ for all } x; \\ c(\boldsymbol{\theta}) &= c(\theta_1, \theta_2) = \exp\left(-\frac{\theta_2^2}{2\theta_1}\right), \quad (\theta_1, \theta_2) \in (0, \infty) \times \mathbb{R} \\ w_1(\boldsymbol{\theta}) &= \theta_1 & t_1(x) &= -x^2/2 \\ w_2(\boldsymbol{\theta}) &= \theta_2 & t_2(x) &= x \end{aligned}$$

Therefore, by the above theorem

$$\begin{aligned} E(X) &= -\frac{\partial}{\partial \theta_2} \log c(\boldsymbol{\theta}) = \frac{\theta_2}{\theta_1} = \mu \\ \text{Var}(X) &= -\frac{\partial^2}{\partial \theta_2^2} \log c(\boldsymbol{\theta}) = -\frac{1}{\theta_1} = \sigma^2 \end{aligned} \tag{2}$$

Joint and Marginal Distributions

In previous lectures, we have discussed probability models and computation of probability for events involving only one random variable. These are called *univariate models*.

In an experimental situation, it would be very unusual to observe only the value of one random variable. For example, in an experiment designed to gain information about some health characteristics of a population of people, the body weights of several people in the population might be measured. These different weights would be observations on different random variables, one for each person measured. Multiple observations could also arise because several physical characteristics were measured on each person. Thus, we need to know how to describe and use probability models that deal with more than one random variable at a time.

Definition: An n -dimensional random vector $\mathbf{X} = (X_1, \dots, X_n)$ is a function from a sample space S into $\mathbb{R}^n$.

- Each coordinate X_i is a random variable.
- The random vector is associated with a probability space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), F)$.
- For each Borel set B ,

$$\Pr\{\mathbf{X} \in B\} = \Pr\{\mathbf{X}^{-1}(B)\} \tag{3}$$

where

$$\mathbf{X}^{-1}(B) = \{w : \mathbf{X}(w) \in B\}$$

Example (Bivariate random variable) A fair coin is flipped 3 times. Define the random vector (X, Y) where X represents the number of heads on the last toss and Y the total number of heads. Then, the probabilities of various outcomes are given in the following table:

Outcome	(x, y)	$\Pr(outcome)$
(H, H, H)	(1, 3)	1/8
(H, T, H), (T, H, H)	(1, 2)	2/8
(H, H, T)	(0, 2)	1/8
(T, T, H)	(1, 1)	1/8
(T, H, T), (H, T, T)	(0, 1)	2/8
(T, T, T)	(0, 0)	1/8

Definition Two random variables X and Y are said to be jointly *discrete* if there is an associated *joint probability mass function*,

$$f_{X,Y}(x, y) = \Pr\{X = x, Y = y\}$$

which sums to 1 over a finite or possibly countable combinations of x and y for which $f_{X,Y}(x, y) > 0$, i.e.,

$$\sum_{x,y} f_{X,Y}(x, y) = 1$$

From this, one can also obtain the marginal pmfs of X and Y as follows:

$$f_X(x) = \Pr(X = x) = \sum_y f_{X,Y}(x, y)$$

$$f_Y(y) = \Pr(Y = y) = \sum_x f_{X,Y}(x, y)$$

Example Back to the fair coin example again. From the definition, we can construct the joint pmf of X and Y :

		Y			
		0	1	2	3
X	0	1/8	1/4	1/8	0
	1	0	1/8	1/4	1/8

The marginal distributions of X and Y are also easy to find. Note: Marginals do not determine joint pmf.

Bivariate cdfs Whether they are discrete or continuous or some combination of the two, we can always define the *joint cdf*. For $n = 2$, the *bivariate cumulative distribution function* is

$$F_{X,Y}(x, y) = \Pr\{X \leq x, Y \leq y\}$$

Properties:

- $F_{X,Y}(x, y) \geq 0$

- $F_{X,Y}(\infty, \infty) = 1$
- $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$
- $F_{X,Y}(-\infty, -\infty) = 0$
- F is non-decreasing and right-continuous in each variable separately.

Joint probabilities All joint probability statements about X and Y can be answered in terms of their joint cdf:

$$\Pr(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{X,Y}(x_2, y_2) + F_{X,Y}(x_1, y_1) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1)$$

Example

$$\Pr(X > x, Y > y) = 1 - F_X(x) - F_Y(y) + F_{X,Y}(x, y)$$

Note: To ensure that a bivariate function $F(x, y)$ is a proper cdf, it must satisfy all the properties mentioned above and the rectangular property above.

Marginal distributions From $F_{X,Y}$, we can derive the univariate distribution functions for X and Y . These are generally called *marginal distributions*.

$$\begin{aligned} F_X(x) &= \Pr\{X \leq x\} = \Pr\{X \leq x, Y \leq \infty\} = F_{X,Y}(x, \infty) \\ F_Y(y) &= \Pr\{Y \leq y\} = \Pr\{X < \infty, Y \leq y\} = F_{X,Y}(\infty, y) \end{aligned}$$

Note: Although we can obtain $F_X(x)$ and $F_Y(y)$ from the joint cdf, we cannot do the reverse.

Continuous Bivariate RVs The random variables X and Y are said to be *jointly continuous* if there exists a function $f_{X,Y}(x, y)$, such that for any Borel set B of 2-tuples in $\mathbb{R}^2$,

$$\Pr\{(X, Y) \in B\} = \int \int_{(x,y) \in B} f_{X,Y}(x, y) dx dy.$$

The function $f_{X,Y}(x, y)$ is called the *joint probability density function* for X and Y . It follows in this case that

$$\begin{aligned} F_{X,Y}(x, y) &= \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) dt ds, \\ f_{X,Y}(x, y) &= \frac{\partial^2 F(x, y)}{\partial x \partial y} \end{aligned}$$